

Comparison Between Vector Recording and TDMR: Inter-track Interference, and Write Synchronization Effects

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The increasing demand for data storage in hard disk drives (HDDs) motivates the development of heat-assisted magnetic recording (HAMR) combined with vector recording (VR) as a potential solution for future ultra-high-density storage systems. In this work, the fundamental limits of vector recording are investigated through comparison with two-dimensional magnetic recording (TDMR), with emphasis on inter-track interference, and write synchronization. Thermally induced reader mag noise are modeled micromagnetically, while the inter-track interference coupling coefficient (β) is analytically calculated and incorporated into the readback analysis. The results show that vector recording achieves nearly 3.3 dB higher peak SNR than TDMR by inherently suppressing inter-track interference without aggressive reader scaling. However, down-track transition offsets significantly affect signal superposition, particularly crosstalk near the inter-track boundary in vector recording, where as a small offset can slightly improve the TDMR SNR. These results demonstrate the robustness and signal integrity advantages of vector recording for future high-density HAMR systems.

Index Terms—Areal Density Capacity, Inter-track Interference, HAMR, TDMR, Reader Design, Vector Recording.

I. INTRODUCTION

THE AMOUNT of digital data generated worldwide is increasing rapidly due to technologies such as artificial intelligence and modern communication systems. As a result, the HDD industry is under growing pressure to further enhance areal density capacity (ADC), while existing recording technologies are approaching their scaling limits [1]. To continue increasing storage density, HDD technology has evolved from perpendicular magnetic recording (PMR) to HAMR, where smaller bit lengths and narrower tracks enable higher ADC. However, further scaling toward ultra-high densities requires substantial reader size reduction, which creates major challenges for improving bits per inch (BPI) and tracks per inch (TPI). TDMR, illustrated in Fig. 1(a) with ellipse B, can partially ease these reader scaling requirements while improving ADC [1], [2]. Nevertheless, as recording density increases, reader noise and inter-track interference become dominant factors limiting the achievable signal-to-noise ratio (SNR), making conventional TDMR insufficient for future HDD demands. In our earlier work, we implemented vector recording, shown in Fig. 1(a) with ellipse A, where one conventional reader and one vector reader are used to detect magnetic field components parallel and perpendicular to the recording layer [1]. This approach improves readback performance while reducing the need for aggressive reader dimension scaling. We also proposed alternative vector reader configurations with free-layer bias angles of 45° and 135°, which achieved similar performance through the X_1+X_2 and X_1-X_2 operations, where X_1 and X_2 represent the readback signals from the 45° and 135° readers, respectively [1].

In our previous study [1], a direct performance comparison between vector recording and TDMR was not carried out and remains an important open question. It has been argued that TDMR using two reduced-size conventional readers can approach vector recording performance through the $S_2-\beta \times S_1$ or $S_1-\beta \times S_2$ operations where S_1 and S_2 are the readback signals from two conventional readers [2]. However, the effectiveness of these operations under strong inter-track interference and realistic recording conditions needs further investigations. In addition, the previous work considered perfectly synchronized

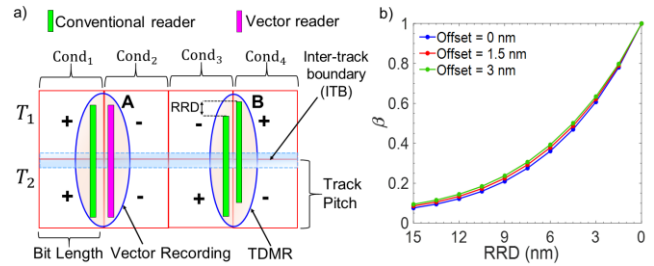


Fig. 1. (a) Readback strategy for vector recording and TDMR. (b) β vs reader-to-reader distance (RRD) for different down-track offsets of Track 2 relative to Track 1.

writing between adjacent tracks, while synchronization mismatch between Track 1 and Track 2 can significantly affect signal superposition, crosstalk, and overall readback performance. In this work, these issues are systematically analyzed using micromagnetic simulations.

II. β CALCULATION

With conventional TDMR, three magnetic states are primarily distinguishable: ++ (cond₁), -- (cond₂), while +- (cond₃) and -+ (cond₄) are considered as a single combined state as shown in Fig. 1(a). Although TDMR can theoretically resolve all four magnetic states using the $S_2-\beta \times S_1$ operation, achieving this requires aggressive reader scaling and accurate estimation of the inter-track coupling coefficient β . As a result, TDMR becomes more sensitive to reader noise and inter-track interference. To estimate the inter-track coupling coefficient β , a pseudo-random bit sequence (PRBS) is written on Track 1 with minimum bit length = 12 nm, while a single-tone (ST) pattern is written on Track 2. The readback signals are modeled using a linear channel with additive noise as

$$S_1(t) = H_1 a_1(t) + G_{12} a_2(t) + n_1(t) \quad (1)$$

$$S_2(t) = G_{21} a_1(t) + H_2 a_2(t) + n_2(t) \quad (2)$$

where H_1 and H_2 represent the on-track responses, G_{12} and G_{21} denote the inter-track coupling terms, and n_1 and n_2 are additive noise components [3]. The channel coefficients are extracted using a least-squares (LS) approach, and the normalized coupling coefficient is defined as $\beta = G_{21}/H_1$, which is evaluated as a function of reader-to-reader distance (RRD) as shown in Fig. 1(b). To investigate the effect of write synchronization on β , an additional down-track offset is introduced on Track 2

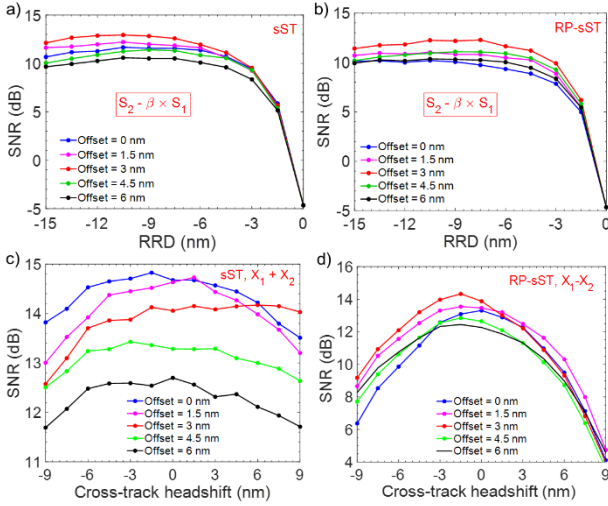


Fig. 2. SNR characteristics for (a) TDMR with sST, (b) TDMR with RP-sST, (c) VR with sST, and (d) VR with RP-sST.

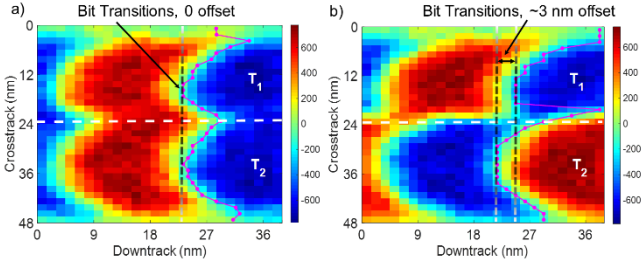


Fig. 3 HAMR transition patterns generated by averaging over 500 bit transitions in the absence of any externally introduced down-track offset on Track 2 (offset = 0 nm) for (a) sST, (b) RP-sST.

relative to Track 1. However, since the β estimation already uses PRBS on Track 1 and a ST pattern on Track 2, an inherent bit transition asynchronization already exists between the tracks. As expected, the additional small down-track offsets produce negligible variation in β , as shown in Fig. 1(b).

III. SNR ANALYSIS

Two recording patterns with bit length (BL) = 20 nm are considered: same-polarity single-tone bits (sST) and reverse-polarity single-tone bits (RP-sST) between adjacent tracks. Fig. 2(a) and 2(b) demonstrate the SNR performance of TDMR with a two-reader configuration (reader width = 21 nm) as a function of RRD for various down-track offsets with sST and RP-sST pattern, respectively. The coupling coefficient β is estimated for each RRD and applied in the $S_2 - \beta \times S_1$ operation during readback. At RRD = 0 nm, both readers are positioned at the center of the inter-track boundary (ITB) region, resulting in maximum adjacent-track interference. As the RRD increases, the interference decreases and the SNR improves, reaching its maximum at RRD = 9 nm to 10.5 nm, where the readers are positioned at the center region of their respective tracks (track pitch = 18 nm). Beyond the optimum RRD, the SNR starts to decrease due to the reduced sensitivity to the target-track signal as the readers move farther apart. Using the $S_2 - \beta \times S_1$ operation, TDMR achieves a maximum SNR of 11.8 dB for the sST pattern and 10.3 dB for the RP-sST pattern when down-track offset value is zero (blue lines in both Fig. 2(a) and 2(b)). In

comparison, vector recording with 45° and 135° readers using the $X_1 + X_2$ and $X_1 - X_2$ operations achieves peak SNR values of 14.83 dB and 13.3 dB for the sST and RP-sST patterns, respectively, as shown in Fig. 3(c) and (d) with offset = 0 nm (blue lines). Therefore, vector recording provides nearly 3 dB higher SNR than TDMR without relying on aggressive reader scaling and inter-track interference cancellation.

IV. EFFECTS OF WRITE SYNCHRONIZATION

From Fig. 2(a) and 2(b), the TDMR SNR initially increases with down-track offset, reaches a maximum with the offset of 3 nm, and then decreases with further rise of the offset value. For the sST case, the bit transitions are initially synchronized at 0 nm offset, as shown in Fig. 3(a), and the adjacent-track signals possess the same polarity because only the ++ and -- states are present. Consequently, the $S_2 - \beta \times S_1$ operation becomes naturally subtractive. Introducing a small down-track offset slightly shifts the transition waveforms along the down-track direction and modifies their phase relationship, making the operation more additive and slightly improving the SNR. For the RP-sST case, the adjacent-track signals possess opposite polarity, so the $S_2 - \beta \times S_1$ operation is naturally more additive. A small offset further compensates for transition curvature and improves local transition alignment as an inherent ~ 3 nm transition shift exists even under synchronized writing condition (offset = 0 nm) due to the curved bit transition as shown in Fig. 3(b), maximizing the resultant signal amplitude. However, larger offsets in both cases reduce waveform correlation and distort the target-track response, increasing residual crosstalk and lowering the SNR.

From Fig. 2(c), the VR SNR decreases as the offset increases because the temporal overlap of the ++ and -- magnetic field components is reduced, weakening constructive signal superposition. In contrast, from Fig. 2(d), the SNR initially increases, reaches a maximum near a 3 nm offset, and then decreases with further offset. This behavior originates from curved bit transitions as shown in Fig. 3(b). Therefore, introducing a 3 nm down-track offset improves transition alignment and maximizes the SNR, highlighting the strong sensitivity of vector recording to write synchronization. In summary, a small down-track offset can slightly improve TDMR SNR by enhancing inter-track interference cancellation through the $S_2 - \beta \times S_1$ operation. In vector recording, write synchronization is critical, where small down-track offsets can also slightly improve SNR depending on transition curvature and polarity configuration, but simultaneously increase reader crosstalk.

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